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Oefeningenreeks 2E nr. 10.10

$$2^{2x+1} - 9 \cdot 2^{x-1} + 1 = 0$$

Substitueer  $2^x = y$

$$\Rightarrow y^2 \cdot 2 - 9 \cdot \frac{y}{2} + 1 = 0$$

$$D = b^2 - 4ac$$

$$= \frac{81}{4} - 4 \cdot 2 \cdot 1$$

$$= \frac{81}{4} - 8 \cdot \frac{4}{4}$$

$$= \frac{81 - 32}{4}$$

$$= \frac{49}{4}$$

$$y_1 = \frac{-b - \sqrt{D}}{2a} = \frac{\frac{9}{2} - \frac{7}{2}}{4} = \frac{1}{4}$$

$$y_2 = \frac{-b + \sqrt{D}}{2a} = \frac{\frac{9}{2} + \frac{7}{2}}{4} = \frac{8}{4} = 2$$

Omzetten naar  $x$ :

$$y = \frac{1}{4} \Rightarrow 2^x = \frac{1}{4}$$

$$\Rightarrow 2^x = \frac{1}{2^2}$$

$$\Rightarrow x = -2$$

$$y = 2 \Rightarrow 2^x = 2$$

$$\Rightarrow x = 1$$

$$V = \{-2, 1\}$$

## References